

Android App Laboratory Activity

Objective

Develop an Android application that demonstrates basic concepts of Android development and data structures. The application should allow users to compute the average of grades and manage a list of contacts.

App Overview

The app consists of two main functionalities:

1. Compute Average
2. Manage Contacts

Screens and Features

1. Main Page

- **Description:** This is the landing page of the app.
- **Components:**
 - **TextView:** Displays "Android Use Case".
 - **Button:** "Compute Average" to navigate to the Compute Average screen.
 - **Button:** "Contacts" to navigate to the Contacts List screen.

2. Contact List

- **Description:** Displays a list of contacts and allows adding new contacts.
- **Components:**
 - **TextView:** Displays "Contact List".
 - **Button:** "Add Contact" to navigate to the Add Contact screen.
 - **ListView:** Displays a list of contacts with their names and details.

3. Contact Details

- **Description:** Displays detailed information of a selected contact with options to mimic a call or delete the contact.
- **Components:**
 - **TextView:** Displays contact's name, phone number, email, and age.
 - **ImageView:** Icons for call and delete actions.
 - **Button:** Back button to return to the Contact List.

4. Add Contacts

- **Description:** Allows users to add a new contact.

- **Components:**

- **EditText:** Input fields for Name, Contact Number, Email, and Age.
- **Button:** "Submit" to add the contact to the list.
- **Button:** "Back" to return to the Contact List.

5. Compute Average

- **Description:** Allows users to input grades for different subjects and compute the average.
- **Components:**

- **EditText:** Input fields for grades in Math, Filipino, Science, MAPEH, and English.
- **Button:** "Submit" to compute the average.
- **Button:** "Back" to return to the Main Page.

6. Average Result

- **Description:** Displays the computed average of the grades.
- **Components:**

- **TextView:** Displays the computed average.
- **Button:** "Back" to return to the Compute Average screen.

Instructions

1. Setup and Initialization:

- Set up your Android development environment (Android Studio).
- Create a new project with an Empty Activity.

2. Design the Layouts:

- Design the layouts as shown in the images using XML.
- You can use different Layout for flexible and adaptive UI.
- Ensure proper alignment and spacing of components.

3. Implement Navigation:

- Implement navigation between different screens using Intents.

4. Manage Data:

- Use appropriate data structures (e.g., ArrayList) to manage the list of contacts.
- Store and retrieve contact details.

5. Compute Average:

- Implement the logic to compute the average of the grades inputted by the user.
- Display the result in the Average Result screen.

6. Add Functionalities:

- **Input Validation:** Ensure that the user inputs are valid (e.g., non-empty fields, valid email format, age within a reasonable range).
- **System Messages:** Display appropriate messages for actions like adding a contact, deleting a contact, or computing the average.
- **Unique Special Functionality:** Add one unique feature to the app. This could be anything that enhances the app's functionality, such as:
 - Exporting contact list to a file.
 - **DO NOT APPLY DATABASE CONCEPT HERE.**

Evaluation Criteria

- **Layout Design:** Proper use of XML to create user interfaces.
- **Navigation:** Smooth navigation between different screens.
- **Functionality:** Correct implementation of features and functionalities.
- **Data Management:** Efficient use of data structures.
- **Input Validation:** Proper validation of user inputs.
- **System Messages:** Informative and user-friendly messages.
- **Creativity:** Implementation of a unique feature.

Grading Rubric

Grade Range	Description	Remarks
70 - 74	Failed	Needs significant improvement
75 - 80	Satisfactory	Basic understanding
81 - 85	Good	Proficient
86 - 90	Very Good	Strong understanding
91 - 95	Excellent	Outstanding performance
96 - 100	Superior	Exceptional mastery

Descriptions and Remarks

- **70 - 74: Failed**
- **Description:** The student has not met the basic requirements and shows a lack of understanding.
- **Remarks:** Needs significant improvement in understanding and application of the concepts.
- **75 - 80: Satisfactory**

- **Description:** The student has a basic understanding of the material but may have some gaps.
- **Remarks:** Demonstrates basic proficiency but requires further practice and study.

- **81 - 85: Good**

- **Description:** The student shows proficiency in the subject with a good grasp of the material.
- **Remarks:** Proficient but there is room for improvement to achieve a higher level of mastery.

- **86 - 90: Very Good**

- **Description:** The student has a strong understanding of the material and performs well.
- **Remarks:** Shows strong proficiency with minor areas for improvement.

- **91 - 95: Excellent**

- **Description:** The student demonstrates outstanding performance and a thorough understanding of the concepts.
- **Remarks:** Performs at a high level with a deep understanding of the material.

- **96 - 100: Superior**

- **Description:** The student exhibits exceptional mastery of the subject matter and exceeds expectations.
- **Remarks:** Demonstrates exceptional understanding and application, with no significant areas for improvement.